

Two arms with equal budgets and a blinded evaluator

Islands arm: agents share findings

Temporal

Composition

Interactions

6 proposals per agent, 3 repetitions

reports after proposals 2 and 4 (at most 120 words)

Independent arm: control, no sharing

Temporal

Composition

Interactions

6 proposals per agent, 3 repetitions

no communication

proposals in, feedback out

LLM judge: did a later proposal use a reported claim?

Blinded evaluator, research data only
anonymous variables, groups and months
feedback: IC, t-stat, coverage, correlation

Repetitions 2 and 3
measure the search only;
they never feed a strategy

Human audit of
judged pairs
(17 in the Opus study)

Mechanical selection, repetition 1 only
 $t \geq 1.5$, same sign in both halves,
 $|\text{corr}| < 0.8$, at most 3 features per arm

A4 = ridge on F1-F5
+ island-arm features

A3 = ridge on F1-F5
+ independent-arm features